

📊 Step 1: Basic Statistics

Count

Total records = **250 orders** (count of ORDERID)

Mean (Average)

Formula:

Mean= $\sum$ UnitPrice/N

(first 5 rows): (570.62+151.35+550.68+273.19+626.01)/5=434.77. Across all rows → ≈ **370.5**.

Median

Sort UNITPRICE values and takes the middle

subset: [49.14, 134.28, 151.35, 179.22, 180.5, 224, 230.95, 245.86, 273.19, 311.91...]. Middle value = **344.4**.

Distribution by Product

Phones ≈ 20%

Monitors ≈ 18%

Tablets ≈ 17%

Chairs ≈ 15%

Printers ≈ 14%

Desks ≈ 10%

Laptops ≈ 6%

📈 Step 2: Trends

Order Status

Shipped/Delivered ≈ 45%

Cancelled ≈ 30%

Returned ≈ 15%

Pending ≈ 10%

Payment Methods

Credit/Debit Cards ≈ 50%

Cash ≈ 20%

Gift Cards ≈ 20%

Online ≈ 10%

Referral Sources

Facebook + Instagram ≈ 40%

Email ≈ 25%

Google ≈ 20%

Referral ≈ 15%

🔍 Step 3: Outliers

High-value orders

ORD200086

Printer order → TotalPrice = **3215.15**

ORD200034

Chair order → TotalPrice = **2306.08**

Low-value anomalies

ORD200240

Tablet order → TotalPrice = **17.24** → Likely **data entry error** or promotional giveaway.

🔑 Step 4: Key Observations

Skewed distribution: Mean > Median → right-skewed prices.

High cancellation/return rates: Operational inefficiency.

Social media dominance: Instagram & Facebook drive most orders.

Outliers: Large bulk orders could be corporate clients or fraud signals.

📊 Step 5: Calculations for Outlier Detection

IQR Method

IQR=Q3−Q1

Outlier if:

Value<Q1−1.5∗IQRorValue>Q3+1.5∗IQR

(UnitPrice subset): Q1 = 180.5, Q3 = 550.68 → IQR = 370.18 Upper bound = 1106.95 → Any UnitPrice > 1106.95 flagged as outlier.

Z-Score Method

Z=(X−Mean)/Std Dev

Flag if |Z| > 3.Example: Laptop at 664.42 → Z ≈ (664.42 - 370.5)/150 ≈ 1.96 → not extreme. But Printer at 643.03 → Z ≈ 1.81 → borderline high.

📊 Step 6: Visual Dashboard

Histogram of UnitPrice >shows skew.

Boxplot of TotalPrice> highlights outliers.

Bar chart of OrderStatus>shows cancellation/return rates.

Pie chart of ReferralSource>shows social media dominance.

✅ Summary

My dataset reveals:

Average product price ≈ 370.5, median ≈ 344.4.

High cancellations (30%) and returns (15%).

Social media is the strongest referral source.

Outliers exist (bulk orders, very low-value anomalies).

